

Dense Supervision, Gentle Decay: Curriculum-Guided Rebalancing of ELECTRA-Style RTD and MLM for Small-Data Pretraining

Aftab Hossain Mikat and Tamjid Hasan Fahim

Department of Computer Science and Engineering
Rajshahi University of Engineering & Technology, Bangladesh
ahmikat@gmail.com, tamjidhfahim@gmail.com

Abstract

When training data is very limited, the way different training signals are weighted over time can strongly affect what a model learns. We study hybrid replaced-token detection (RTD) and masked language modeling (MLM) pretraining under the BabyLM 2026 Strict-Small budget. We train five single-seed DeBERTa-v2 models along three curriculum axes: objective scheduling, masking policy, and data order. The primary system, bgb-decay, retains generator MLM, discriminator RTD, and auxiliary discriminator MLM throughout training while reducing the RTD coefficient in stages. We compare it with a matched fixed-RTD reference, a variant that adds easy-to-hard data ordering and adaptive auxiliary masking, and two stronger RTD-to-MLM handoff schedules with whole-word and adaptive masking. Among the five runs, bgb-decay achieves the best overall balance, while the more aggressive variants gain more on some tasks but drop more on others. Training diagnostics reveal a trade-off between replacement discrimination and auxiliary vocabulary prediction: under fixed high RTD weighting, RTD gradients are larger and nearly orthogonal to auxiliary-MLM gradients, while staged decay preserves RTD but shifts late optimization toward MLM. These results suggest that gradual objective rebalancing is a promising direction for small-data pretraining.

1 Introduction

The BabyLM Challenge studies language-model pretraining under severe data constraints, with the goal of understanding which learning strategies remain effective when models cannot rely on web-scale corpora (Warstadt et al., 2023). In the Strict-Small track, training is limited to a 10-million-word corpus and at most ten passes over that corpus (BabyLM Organizers, 2026). Under this constraint, training design becomes especially important: the same text can provide very different learn-

ing signals depending on which objective dominates, which tokens are selected for prediction, and which examples are presented first. We study this question through a hybrid pretraining setup that combines masked language modeling (MLM) with replaced-token detection (RTD) rather than treating them as competing objectives; Section 2 situates this choice relative to prior BabyLM systems and pretraining objectives.

Most curriculum-learning work in this space varies a single axis, such as example order or masking difficulty, while holding the underlying objective fixed (Schoenegger et al., 2025; Lucas et al., 2024). We instead organize our experiments along three curriculum axes that can be varied independently or in combination. An **objective curriculum** changes which losses are active or how strongly they are weighted. A **corruption curriculum** changes what the model must predict, for example by moving from uniform token masking to whole-word or difficulty-adaptive masking policy. A **data curriculum** changes when training examples are presented, such as ordering examples from easier to harder before returning to random shuffling. This distinction provides a common structure for the five runs in our study and makes it possible to compare otherwise heterogeneous interventions on equal footing.

Our main model, **bgb-decay**, varies only the objective axis: it keeps RTD and MLM jointly active for the full training budget but stages the RTD coefficient downward over time. We compare it against a matched fixed-RTD reference, a curriculum extension of that reference, and two more aggressive schedules that fully hand off from RTD to MLM partway through training, three of which were additionally motivated by improving performance on Human-like evaluation metrics such as Age of Acquisition (AoA) (Fenson et al., 2007; Chang and Bergen, 2022). Across the five single-seed runs, bgb-decay achieves the strongest aggregate profile,

and its advantage is broad rather than concentrated in one task family: the more aggressive variants post the best scores on individual categories but also the largest losses elsewhere. This pattern suggests the central question is not simply whether to use RTD, MLM, or curriculum learning, but how to allocate a limited optimization budget among these signals over the course of training.

2 Related Work

Data-constrained pretraining and shared artifacts. The BabyLM Challenge asks which pre-training strategies remain effective when a model is limited to a developmentally plausible amount of text rather than a web-scale corpus (Warstadt et al., 2023; Hu et al., 2024; Choshen et al., 2026). We use the organizers’ 2026 Strict-Small corpus release (BabyLM Organizers, 2026), evaluation implementation, and leaderboard. Prior BabyLM submissions vary architecture, data, and optimization as well as curriculum. Our five runs instead share the corpus, tokenizer family, backbone dimensions, and budget. This improves comparability, although only bgb-decay versus Baby Got BERT is a one-factor comparison; the other runs remain system-level comparisons.

Replaced-token detection versus masked language modeling. ELECTRA introduced replaced-token detection (RTD) as an alternative to MLM: a small generator proposes replacement tokens, and a discriminator is trained to detect which tokens in the resulting sequence were replaced (Clark et al., 2020). Because RTD supervises every non-special token rather than only a masked subset, it has been argued to be a more sample-efficient objective than MLM, a property that is directly appealing under the Strict-Small budget. We adopt a compact DeBERTa-v2 encoder as our discriminator backbone (He et al., 2021b,a). Rather than treating RTD and MLM as competing choices of pretraining objective, as is typical when comparing ELECTRA-style and BERT-style models, we combine them in a single discriminator and study how the *balance* between them should change over training — a question that sits one level above the usual RTD-versus-MLM comparison.

Data curricula. Curriculum learning conventionally orders or selects examples from easier to harder (Bengio et al., 2009); competence-based variants

gradually expand the portion of data available to the learner (Platanios et al., 2019). BabyLM work has instantiated difficulty using sentence complexity (Oba et al., 2023), model uncertainty (Hong et al., 2023), acquisition-theoretic stages (Salhan et al., 2024), and training-example influence (Schoeneger et al., 2025). These studies report mixed, often capability-specific outcomes, so “curriculum” is not a single intervention with a single expected effect. Our *data curriculum* label is reserved for this example-ordering family; LAD is our only run that changes this axis.

Corruption and objective curricula. Curricula can also change the prediction problem without re-ordering examples. Whole-word masking (WWM) groups all subword pieces of a lexical item into one masking unit (Cui et al., 2021), while adaptive masking emphasizes tokens the current model predicts poorly (Yang et al., 2023; Edman and Fraser, 2025). Developmentally staged masking has also been explored in BabyLM by expanding the lexical or part-of-speech classes eligible for masking (Lucas et al., 2024). We call these *corruption curricula* because they change which targets are predicted. In contrast, our *objective curriculum* changes the relative weight or availability of RTD and MLM. This three-axis terminology is the paper’s main conceptual distinction: prior example-order and masking curricula motivate LAD and Lilliput, while the matched bgb-decay comparison tests whether the balance between dense discrimination and lexical prediction should itself be scheduled.

3 Training Framework and Curriculum Axes

3.1 Hybrid Dense and Lexical Supervision

With a standard 15% mask rate, MLM applies its vocabulary-prediction loss only to the masked positions; unmasked tokens still provide context but are not themselves prediction targets. RTD instead turns most of the sequence into supervised signal: a generator fills selected positions with sampled replacements, and the discriminator predicts whether each non-special token in the resulting sequence is original or replaced. The two objectives are therefore related but not equivalent. RTD asks whether an observed token fits its context, while MLM asks which vocabulary item belongs in a missing position. So our systems train the discriminator with both signals. For a clean sequence x , the generator predicts masked tokens and produces a corrupted

sequence $\tilde{x}$; the discriminator then performs RTD on $\tilde{x}$ and vocabulary prediction on a separately masked version of the clean sequence. The joint loss at step t is

$$\mathcal{L}_t = \mathcal{L}_{\text{G-MLM}} + \lambda_t \mathcal{L}_{\text{RTD}} + \mathcal{L}_{\text{D-MLM}}, \quad (1)$$

where the generator MLM and discriminator MLM coefficients are fixed at one and λ_t controls the relative strength of RTD.

This formulation lets us study an important small-data question: *should dense RTD remain the dominant signal for the entire budget, or should its influence decrease once the model has already learned a useful replacement-discrimination representation?* The objective-curriculum experiments vary λ_t , or remove RTD at different stages, to test this trade-off.

3.2 Three Forms of Curriculum

Curriculum learning can intervene at different points in the training pipeline, so we distinguish three forms.

Objective curriculum. An objective curriculum changes the active losses or their relative weights: our schedules range from fixed high RTD, to staged RTD decay, to a complete transition from joint RTD+MLM training into discriminator-only MLM. This axis directly controls how strongly dense discrimination competes with vocabulary prediction in the shared encoder; complete checkpoint-level monitor-loss trajectories appear in Appendix D (Figure 4).

Corruption curriculum. A corruption curriculum changes the prediction problem itself. *Uniform token masking* samples eligible positions without considering model difficulty; *whole-word masking* (WWM) instead treats a lexical word as one masking unit, so all its subword pieces are masked together (Cui et al., 2021); and *adaptive WWM* assigns greater masking probability to words or word classes the model currently predicts poorly, directing more updates toward difficult lexical targets (Yang et al., 2023; Edman and Fraser, 2025). Our adaptive variants retain a uniform exploration component so that easier or less frequently selected words remain represented in training. The exact weighting rule is given in Appendix B.

Lilliput Curriculum additionally uses a *short-context bias* during its first stage, giving greater masking priority to utterances of at most five

whitespace-delimited words while a uniform component keeps longer contexts represented, a coarse developmental prior testing whether locally constrained contexts before harder lexical targets could improve acquisition-oriented behavior, especially AoA.

Data curriculum. A data curriculum changes the order in which examples are seen. LAD Got BERT assigns each packed sequence a corpus-derived difficulty score based on lexical rarity, subword fragmentation, sentence length, and insufficient context; training begins with a stronger easy-to-hard order, weakens that preference over the next two epochs, and becomes uniformly shuffled from epoch 4 onward. Every example is still seen once per epoch, so only its presentation order changes. The full scoring function appears in Appendix C.

These axes affect different parts of learning: data curriculum controls *when* an example is seen, corruption curriculum controls *what must be predicted* within it, and objective curriculum controls *which learning signal has the most influence* on the shared representation. This separation is central to our analysis, a sophisticated masking or data policy still operates inside whatever objective balance the training procedure imposes.

4 Experimental Setup

4.1 Shared Data, Tokenization, and Backbone

All runs use the official BabyLM 2026 Strict-Small corpus (BabyLM Organizers, 2026), containing 10 million whitespace-counted words in the pinned training revision. No external teacher model, synthetic data, evaluation examples, AoA labels, or other evaluation labels are used for training, curriculum ordering, or adaptive mask selection. The maximum budget is ten corpus passes, represented as 100M cumulative word presentations. Nineteen milestone artifacts are exported: 1M through 9M, followed by 10M through 100M in 10M increments.

All five runs use a 16K SentencePiece unigram tokenizer (Kudo and Richardson, 2018) trained only on the permitted corpus and pack text into sequences of up to 128 tokens. The discriminator is a 12-layer DeBERTa-v2 encoder with hidden size 384, feed-forward size 1536, and six attention heads. A four-layer generator shares token embeddings with the discriminator. Joint-training phases use micro-batches of 8 with 16-step gradient accumulation, a peak learning rate of 5×10^{-4} , weight

decay 0.01, and seed 42. MLM-only phases use a learning rate of 2×10^{-4} . Full architecture and optimization details are provided in Appendix A.

4.2 Experimental Variants

Table 1 summarizes the five executed schedules. We describe each run by the intervention it makes and the hypothesis that motivated it.

bgb-decay: retain RTD, reduce its late dominance. The main model keeps generator MLM, discriminator RTD, and discriminator MLM active throughout all 100M exposures. Data order remains uniformly shuffled and corruption remains uniform 15% token masking. Only the RTD coefficient changes: 50 from 0–40M exposures, 25 from 40–70M, and 10 from 70–100M. The motivation is to exploit dense discrimination strongly while the encoder is forming general representations, then reduce its relative influence so that vocabulary prediction can contribute more strongly during later refinement. This is a gradual rebalancing, not an objective handoff: RTD is never removed.

Baby Got BERT: fixed high RTD reference. Baby Got BERT uses the same three-objective training structure, uniform 15% masking, and uniform data order as bgb-decay, but keeps the RTD coefficient fixed at 50 for the full budget. This provides the cleanest comparison for the objective-curriculum hypothesis. The question is whether the strong RTD pressure that is useful early in training should remain equally strong after the discriminator has already learned the replacement task.

LAD Got BERT: Baby Got BERT plus curriculum. LAD is deliberately close to Baby Got BERT on the objective axis: generator MLM, RTD, and discriminator MLM remain active throughout training with fixed RTD weight 50. It adds curriculum around that same objective mixture. The data schedule starts with an easy-to-hard ordering in epoch 1, uses weaker difficulty bias in epochs 2 and 3, and returns to uniform shuffling from epoch 4. This tests whether changing the order of evidence can improve learning without changing objective weights.

LAD also adds a later corruption curriculum only to the auxiliary discriminator MLM stream. Epoch 5 is used to collect token-level auxiliary-MLM accuracy statistics. Beginning in epoch 6, these statistics determine which MLM targets receive more emphasis: words that were predicted less accu-

rately receive higher masking probability, while uniform exploration preserves coverage. Generator masking and RTD corruption remain uniform at 15%, so the adaptive policy does not change the replacement task itself. The original motivation was partly acquisition-oriented: first present easier examples, then use the model’s own learned difficulty estimates to focus lexical practice on words that remain hard. LAD therefore asks whether data and target curricula can improve the fixed-RTD reference without changing its objective balance.

Lilliput Lexicon: early RTD scaffold followed by lexical curriculum. Lilliput Lexicon makes a stronger change to both the objective and corruption axes. The first 10M exposures use joint generator MLM, RTD, and discriminator MLM with RTD weight 10. The generator and RTD objective are then removed. From 10–20M, the discriminator trains with uniform WWM at a 20% mask target. From 20–100M, training uses adaptive WWM, with the global mask target gradually reduced from 20% to 15% and 30% of mask selection kept uniform for exploration.

The hypothesis was that RTD might be most useful as an early representation-building scaffold. After that scaffold is established, the remaining budget is devoted to complete lexical units and increasingly difficult words. This broad-to-hard-word progression was also intended to test whether a more acquisition-like lexical curriculum could improve the Human-like evaluation, especially AoA.

Lilliput Curriculum: smoother RTD introduction and withdrawal. Lilliput Curriculum combines the developmental motivation with a more gradual RTD transition. During the first 10M exposures, joint training uses RTD weight 10 together with short-context-biased masking. From 10–20M, masking becomes uniform and RTD weight increases to 20. From 20–40M, joint training continues while RTD is annealed from 20 to zero. The final 40–100M exposures use discriminator-only adaptive WWM, with the mask target reduced from 18% to 15% and 30% uniform exploration.

The $10 \rightarrow 20 \rightarrow 0$ schedule was chosen to make both the introduction and removal of RTD smoother. RTD begins at a moderate weight while training is biased toward short contexts, becomes stronger after the model moves to the full context distribution, and is then withdrawn gradually before the long adaptive-MLM phase. The short-context stage was intended as a coarse child-

learning analogy: more constrained contexts first, followed by broader contexts and harder lexical targets. The experiment therefore combines a staged objective curriculum with developmental corruption and a late hard-word MLM phase.

4.3 Evaluation Suite

We use the official BabyLM 2026 Strict-Small evaluation implementation and report both category-level scores and challenge aggregates. Because the suite measures several distinct capabilities, no single category should be treated as a complete measure of language learning.

BLiMP measures zero-shot grammatical preferences using minimal contrasts (Warstadt et al., 2020); BLiMP Supplement is the additional held-out set introduced with the first BabyLM evaluation (Warstadt et al., 2023). EWoK evaluates basic world knowledge (Ivanova et al., 2025), COMPS probes conceptual property knowledge and inheritance (Misra et al., 2023), GlobalPIQA measures physical and culturally situated commonsense reasoning (Chang et al., 2025), and Entity Tracking tests whether a model can maintain entities and their changing attributes across a short discourse (Kim and Schuster, 2023).

The challenge’s (Super)GLUE aggregate draws tasks from GLUE and SuperGLUE (Wang et al., 2018, 2019). Its reported score is the unweighted mean of BoolQ accuracy (Clark et al., 2019), MNLI accuracy (Williams et al., 2018), MRPC F1 (Dolan and Brockett, 2005), MultiRC accuracy (Khashabi et al., 2018), QQP F1 as distributed with GLUE, RTE accuracy (Dagan et al., 2006), and WSC accuracy (Levesque et al., 2012). GlobalPIQA is the macro-average of its parallel and nonparallel English subsets, evaluated using length-normalized conditional likelihoods. Reading measures the additional variance in human reading behavior explained by model surprisal after controlling for lexical and contextual predictors (Frank et al., 2013; de Varda et al., 2024). Its category score is the mean of the normalized change in R^2 for self-paced reading, including one-word spillover, and the corresponding score averaged across four eye-tracking measures. AoA fits acquisition curves to word surprisals across training checkpoints and compares the resulting model ages of acquisition with child estimates derived from MacArthur–Bates CDI norms (Fenson et al., 2007; Frank et al., 2017; Chang and Bergen, 2022). Under the current evaluator, AoA is the model–child Pearson correlation,

reported only when it passes the scorer’s significance criterion and otherwise set to zero.

The NLP Average is the unweighted mean of BLiMP, BLiMP Supplement, EWoK, Entity Tracking, COMPS, GlobalPIQA, and (Super)GLUE. The Human-like Average is the mean of Reading and AoA, while Overall is the mean of all nine category scores. Because these metrics differ in scale, difficulty, and variance, we use the aggregates only as summaries and interpret the full task profile when comparing training strategies.

Checkpoint-level fast-subset trajectories are reported only as diagnostics in Appendix E; all model comparisons use the final full-suite scores.

5 Results and Analysis

Table 3 reports the complete official result for all five runs. We treat single-seed differences as suggestive rather than statistically established.

5.1 Answers at a Glance and Comparison Logic

The short answer to “*does curriculum help?*” is: *it depends on which curriculum and which outcome*. Table 2 states the valid comparison for each question before we inspect individual tasks. Only the first row changes one factor. The remaining rows change two or more factors and therefore characterize complete training regimes rather than isolating a mechanism.

bgb-decay has the best Overall (40.16) and NLP Average (51.04), but it does not win every task. It leads BLiMP, Entity Tracking, and COMPS; Liliput Lexicon leads EWoK and GlobalPIQA; Liliput Curriculum leads (Super)GLUE and Reading; LAD leads BLiMP Supplement. So bgb-decay’s advantage is about breadth, not dominance: it avoids the large losses that the more specialized schedules take on elsewhere.

Table 3 also lists three baselines from the official 2026 leaderboard. bgb-decay beats the GPT-2 and Interaction baselines on NLP Average, but those systems differ in architecture and training, so the comparison cannot isolate the effect of RTD decay. We leave Baseline-gpt-bert-masked-focus (mntp) out of the aggregate comparisons, since its zero scores on Entity Tracking, COMPS, and GlobalPIQA look like missing evaluations rather than true zeros (Appendix F).

AoA is 0 for all five of our runs, because none of their model–child correlations pass the scorer’s sig-

System	Training schedule	Curriculum intervention and intended test
bgb-decay	0–40M: joint, $\lambda = 50$; 40–70M: joint, $\lambda = 25$; 70–100M: joint, $\lambda = 10$. Uniform 15% token masking and uniform shuffle throughout.	Objective curriculum only. Test whether strong early RTD can be retained but gradually downweighted so that MLM gains relative influence later.
Baby Got BERT	0–100M: joint, fixed $\lambda = 50$. Uniform 15% token masking and uniform shuffle throughout.	Matched fixed-objective reference for bgb-decay. Test sustained high RTD pressure with no data or corruption curriculum.
LAD Got BERT	0–100M: joint, fixed $\lambda = 50$. Easy-to-hard ordering in epoch 1, weaker bias in epochs 2–3, uniform from epoch 4. Epoch 5 collects auxiliary-MLM accuracy; epochs 6–10 use adaptive masks only for auxiliary MLM.	Baby Got BERT plus data and auxiliary corruption curricula. Test whether acquisition-oriented ordering and adaptive lexical practice help without changing objective balance.
Lilliput Lexicon	0–10M: joint, $\lambda = 10$; 10–20M: discriminator MLM with uniform WWM at 20%; 20–100M: adaptive WWM with 20%→15% mask target and 30% uniform exploration.	Early RTD scaffold followed by increasingly difficult lexical prediction. Test whether a broad-to-hard-word progression improves lexical and acquisition-oriented behavior.
Lilliput Curriculum	0–10M: joint $\lambda = 10$ with short-context bias; 10–20M: joint $\lambda = 20$ with uniform masks; 20–40M: joint $\lambda : 20 \rightarrow 0$; 40–100M: adaptive WWM at 18%→15% with 30% exploration.	Smooth RTD introduction and withdrawal, combined with short-context-first and later hard-word training. Test a more gradual developmental RTD-to-MLM curriculum.

Table 1: Executed training schedules. Exposure values are cumulative word presentations over the 10M-word corpus. “Joint” denotes generator MLM, discriminator RTD, and discriminator MLM trained together; λ is the RTD coefficient.

nificance threshold (Appendix F). So the Human-like Average here is really just half of Reading, and we cannot say whether the three acquisition-motivated curricula actually helped AoA.

5.2 Objective Curriculum: Decay Beats a Fixed Weight

bgb-decay and Baby Got BERT are matched on everything except the RTD weight: bgb-decay steps it down $50 \rightarrow 25 \rightarrow 10$, Baby Got BERT keeps it at 50. Decay wins: +0.34 Overall, +0.46 NLP Average, and gains on six of seven NLP categories (BLiMP +0.63, Supplement +0.59, EWoK +0.30, Entity Tracking +0.63, COMPS +0.75, GlobalPIQA +0.43). Fixed RTD only wins on (Super)GLUE (+0.12) and Reading (+0.15). The gaps are small, but the direction is consistent.

The training diagnostics explain why. Generator MLM accuracy barely moves (0.4249 vs. 0.4260), but bgb-decay’s discriminator-MLM accuracy is higher (0.4416 vs. 0.4334) and its replaced-class RTD F1 is lower (0.2397 vs. 0.2757), while coarse RTD accuracy stays about the same (0.9001 vs. 0.9012). This is exactly what the schedule is meant to do: pull weight away from fine-grained replacement discrimination late in training and give more of it to lexical prediction, without ever removing

RTD. The LAD gradient probes (Appendix D) back this up: under a fixed weight of 50, late RTD gradients run about twice the norm of the auxiliary-MLM gradients and point in a nearly orthogonal direction, which is the imbalance decay is correcting.

5.3 Data Curriculum: LAD Shifts Capability, Doesn’t Rebalance

LAD keeps Baby Got BERT’s fixed RTD schedule and adds two things on top: easy-to-hard ordering in epochs 1–3 (Appendix C) and adaptive masking for the auxiliary MLM from epoch 6 (Appendix B). So the right comparison is LAD minus Baby Got BERT, not LAD minus bgb-decay.

Against that reference, LAD loses 0.27 Overall and 0.33 NLP Average. Its one clear win is BLiMP Supplement (+2.85); everything else non-AoA drops, from -2.09 (EWoK) to -0.05 . So the combined LAD curriculum does not help broadly. It buys one grammar-related gain, and not even on the AoA-related behavior it was designed for. Since data order and adaptive masking changed together, we cannot tell which one caused the gain or the losses.

Question	Comparison (first minus second)	Δ Overall	Δ NLP	Observed answer
Does objective rebalancing improve observed performance?	bgb-decay — Baby Got BERT	+0.34	+0.46	Yes in this single-seed comparison: decay improves six of seven NLP categories; the sole NLP exception is (Super)GLUE (−0.12), while Reading also decreases (−0.15).
How does LAD’s combined curriculum compare with fixed RTD?	LAD Got BERT — Baby Got BERT	−0.27	−0.33	It does not improve the aggregate profile: BLiMP Supplement increases by +2.85, whereas every other reported non-AoA category is lower, with differences ranging from −2.09 to −0.05.
How does the early-handoff Lilliput Lexicon regime compare with bgb-decay?	Lilliput Lexicon — bgb-decay	−0.61	−1.00	The regime shows a task-level trade-off: EWoK (+0.89), GlobalPIQA (+2.02), and Reading (+1.48) improve, whereas BLiMP (−3.89) and (Super)GLUE (−3.70) decline substantially.
How does the multi-stage Lilliput Curriculum regime compare with bgb-decay?	Lilliput Curriculum — bgb-decay	−0.18	−0.59	The regime also shows a task-level trade-off: Reading improves by +2.49, with smaller gains on Supplement (+0.42), GlobalPIQA (+0.55), and (Super)GLUE (+0.59), while BLiMP (−2.09), EWoK (−1.61), and Entity Tracking (−1.94) decline.

Table 2: Comparison map for the five systems. Deltas are percentage-point differences computed from the displayed scores in Table 3.

Model	Overall	NLP	Human	BLiMP	Supp.	EWoK	Entity	COMPS	GPIQA	SGLUE	Reading	AoA
bgb-decay	40.16	51.04	2.08	69.82	60.06	51.90	19.61	52.63	34.62	68.65	4.17	0.00
Baby Got BERT	39.82	50.58	2.16	69.19	59.47	51.60	18.98	51.88	34.19	68.77	4.32	0.00
Lilliput Lexicon	39.55	50.04	2.83	65.93	59.69	52.79	18.27	52.06	36.64	64.95	5.65	0.00
Lilliput Curriculum	39.98	50.45	3.33	67.73	60.48	50.29	17.67	52.57	35.17	69.24	6.66	0.00
LAD Got BERT	39.55	50.25	2.09	67.86	62.32	49.51	17.86	51.55	34.14	68.49	4.17	0.00
BabyLM GPT-2 Strict-Small	37.38	48.99	−3.26	65.23	57.25	50.63	19.10	51.81	35.09	63.80	5.63	−12.15
BabyLM Strict-Small Interaction	38.71	49.04	2.55	63.07	58.13	51.45	19.62	51.29	36.14	63.62	5.10	0.00
Baseline-gpt-bert-masked-focus (mntp) [†]	28.49	35.72	3.19	70.36	63.71	49.95	0.00*	0.00*	0.00*	66.02	6.38	0.00

Table 3: Complete official evaluation results for our five runs and three additional models drawn from the official BabyLM 2026 Strict-Small leaderboard. Bold marks the best value within our five-run experiment. The Baseline-gpt-bert-masked-focus (mntp)[†] row’s Entity Tracking, COMPS, and GlobalPIQA scores of 0.00* likely reflect an unevaluated entry rather than true zero performance (Appendix F), so its Overall/NLP Average are not comparable to the other rows. Because AoA is zero in every experimental row, the Human-like Average for our five runs is determined entirely by Reading.

5.4 Corruption Curriculum: RTD Handoffs Trade Breadth for Specialization

Lilliput Lexicon and Lilliput Curriculum both drop RTD entirely and spend most of training on whole-word and adaptive masking instead. Compared to bgb-decay, Lilliput Lexicon gains on EWoK (+0.89), GlobalPIQA (+2.02), and Reading (+1.48), but loses heavily on BLiMP (−3.89) and (Super)GLUE (−3.70); Overall and NLP Average drop by 0.61 and 1.00. That is a trade, not an improvement.

Lilliput Curriculum stays closer to bgb-decay on Overall (−0.18) but is still behind on NLP Average (−0.59). It gets the biggest Reading gain (+2.49), plus smaller ones on Supplement, GlobalPIQA, and (Super)GLUE, while losing on BLiMP, EWoK, and Entity Tracking. Its trade-off is milder than

Lilliput Lexicon’s, but the two runs differ in more than handoff smoothness alone — short-context bias, joint-training length, RTD weight, and mask schedule all change too — so we cannot credit the smoother withdrawal by itself.

One caveat on the auxiliary-MLM accuracies in Table 6: once masking deliberately targets harder words, a lower monitor accuracy can just mean a harder target set, not a weaker model. So we rely on the held-out task scores above rather than raw monitor accuracy here.

5.5 Synthesis

Figure 1 makes the pattern easy to see, and the three axes behave differently. The objective axis (§5.2) is our one matched, single-factor comparison, and it is also the only one that moves the

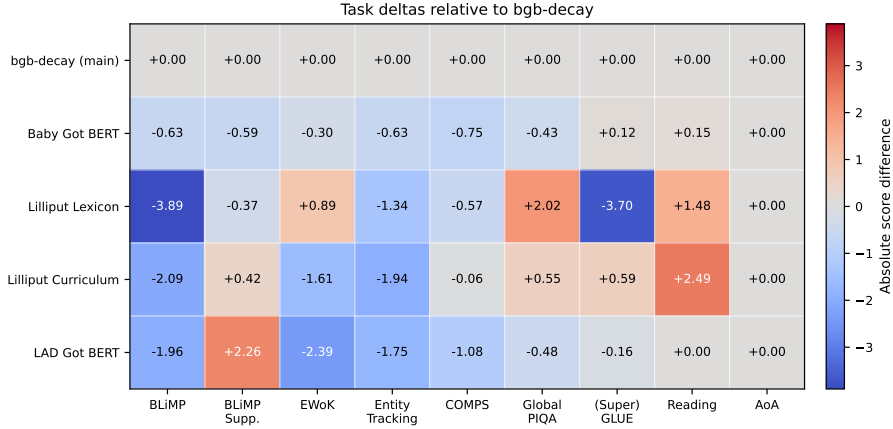


Figure 1: Signed task-score differences relative to bgb-decay. Positive cells indicate that a comparison run has a higher score. The figure emphasizes broad versus localized changes across task families rather than treating each cell as an independent finding.

aggregate broadly instead of locally. The corruption axis (§5.4) does the opposite: both Lilliput variants concentrate their gains and losses into the largest single-category swings in the study, with the smoother handoff softening that trade a little. The data axis (§5.3) sits in between: LAD redirects performance toward one grammatical category without touching the underlying RTD/MLM balance, so it moves things around without broadening them.

One more pattern, worth noting: four of the five runs achieve the highest score within our five-run comparison on at least one category, and each within-study lead aligns with the primary axis modified by that run—bgb-decay on BLiMP, Entity Tracking, and COMPS; LAD on BLiMP Supplement; and the two Lilliput models on the semantic, reasoning, and reading categories where their corruption handoffs concentrate their gains. Baby Got BERT, the only run without an additional curriculum mechanism, does not lead any category.

Our strongest claim stays narrow: between two otherwise-matched joint RTD+MLM runs, reducing RTD after early training gives the better observed balance. The other systems are useful pointers for where to look next, but since each changes several things at once, we cannot say which factor drove its task-specific gains. A natural next step is a factorial design that starts from the decayed schedule and adds data ordering and adaptive masking one at a time, across multiple seeds.

6 Conclusion

We studied five BabyLM Strict-Small pretraining runs through three curriculum axes: objective

scheduling, corruption policy, and data order. This framing turns the runs into a coherent set of comparisons rather than a collection of unrelated submissions. bgb-decay changes only the objective balance, Baby Got BERT provides the matched fixed-RTD reference, LAD adds data and auxiliary masking curricula without changing that fixed objective schedule, and the two Lilliput models test stronger RTD-to-MLM transitions with whole-word and adaptive masking.

The clearest result is on the objective axis. In our matched pair, gradually decaying RTD improves Overall by 0.34 points and NLP Average by 0.46, with gains on six of seven NLP categories. The diagnostics sharpen this: the decayed model has better discriminator-MLM accuracy, while the fixed model keeps a stronger RTD F1. So a better task balance does not require maximizing the RTD metric itself — though multi-seed replication is still needed to call this gain stable.

The other variants do not support a general “curriculum helps” claim. LAD’s combined data-order and masking curriculum improves BLiMP Supplement but hurts both aggregates relative to its fixed-RTD reference. The Lilliput models trade broad performance for gains on specific knowledge, reasoning, reading, or transfer tasks, and each one changes too many factors at once to say why. The natural next step is to keep the decayed objective schedule and add each curriculum axis on its own, under multiple seeds.

Limitations

The study has three main limitations. First, we trained each configuration once, and the five systems were built one after another rather than as a full factorial design. The fixed-vs-decay pair is closely matched, but Lilliput and LAD change several things at once, so their results describe training regimes rather than isolate single causes. We would need multi-seed replication before treating these small score gaps as stable effects.

Second, we settled on bgb-decay only after seeing all five result profiles, and every experiment uses one compact DeBERTa-v2 setup, one corpus, and one training budget. So these findings are hypotheses about objective allocation in this setting, not a universal recipe. The monitor metrics and gradient probes here only support our interpretation; they are not a substitute for held-out evaluation.

Third, the Baseline-gpt-bert-masked-focus (mntp) row in Table 3 comes from the 2026 leaderboard, and its Entity Tracking, COMPS, and GlobalPIQA scores were never re-evaluated under the updated task suite. We treat it as informal context rather than a controlled baseline, and note that it actually beats bgb-decay on BLiMP Supplement and Reading.

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A Architecture and Optimization Details

B Adaptive Whole-Word Masking

Lilliput Lexicon and Lilliput Curriculum maintain prediction counts for token types observed at masked positions. With c_v correct and i_v incorrect predictions, smoothed accuracy is

$$\hat{a}_v = \frac{c_v + 0.5}{c_v + i_v + 1}. \quad (2)$$

A hard-token proposal weight is

$$\tilde{w}_v = \max(w_{\min}, p_t(1 - \hat{a}_v)), \quad (3)$$

where p_t is the current global mask target and $w_{\min} = 0.005$. The implementation updates

$$w_v \leftarrow 0.2w_v + 0.8\tilde{w}_v, \quad (4)$$

normalizes weights to the target average mask rate, caps per-position probability at 0.95, and mixes the adaptive distribution with 30% uniform selection. Whole words are recovered from tokenizer word-group metadata so all subpieces of a selected lexical item are masked together. The exploration component prevents the model from permanently ignoring already-easy words and reduces feedback loops caused by early prediction noise.

LAD Got BERT applies the same general difficulty principle only to the auxiliary discriminator MLM pass. Generator corruption and RTD retain uniform 15% masks. The executed run keeps 16% uniform exploration and updates adaptive weights every 100 optimizer steps. Thus LAD changes the target distribution of one objective without changing the data seen by RTD.

C Corpus-Derived Difficulty Curriculum

LAD computes an ordering score from the training corpus only; the score is removed before the neural forward pass. For normalized word v with frequency f_v and corpus maximum $f_{\max}$,

$$r(v) = 1 - \frac{\log(1 + f_v)}{\log(1 + f_{\max})}. \quad (5)$$

Sentence rarity is the mean of $r(v)$. With k SentencePiece tokens per whitespace word, fragmentation is

$$f = \text{clip}\left(\frac{k-1}{3}, 0, 1\right). \quad (6)$$

For a sentence of n words, the length term is

$$\ell = \min\left(\frac{\log(1 + n)}{\log 31}, 1\right). \quad (7)$$

The insufficient-context penalty $c(n)$ is 1.0 for zero or one word, 0.6 for two, 0.3 for three, and 0 for four or more. The final score is

$$d = 0.45\bar{r} + 0.20f + 0.20\ell + 0.15c(n). \quad (8)$$

Packed-sequence difficulty is the word-count-weighted average of constituent sentence scores. Epoch 1 traverses ten approximately equal-word buckets from easy to hard. Epoch 2 orders by $0.5d + 0.5u$, epoch 3 by $0.2d + 0.8u$, where $u \sim \text{Uniform}(0, 1)$, and epochs 4–10 use a uniform shuffle. Every example appears once per epoch; difficulty controls presentation order, not sampling frequency or model input features.

D Checkpoint-Level Training Diagnostics

The handoff variants make the objective transition explicit: their RTD trajectories end when the generator/discriminator corruption stream is removed. Baby Got BERT, bgb-decay, and LAD retain RTD through 100M exposures, but only bgb-decay reduces its coefficient. bgb-decay maintains nearly the same coarse RTD accuracy as the fixed-RTD reference at the final checkpoint, but its replaced-class F1 is lower. Thus decay preserves the binary task at a broad level while reducing late emphasis on fine-grained replacement discrimination.

The fixed-versus-decay pair is directly comparable because both retain uniform masking. Their final generator MLM accuracies are almost identical, while bgb-decay has higher auxiliary discriminator-MLM accuracy and lower RTD F1. This is the monitor signature expected from reallocation: late capacity shifts toward vocabulary prediction without removing RTD. Absolute comparisons with the adaptive runs are inappropriate because their mask distributions become harder over time.

Checkpoint-level monitor losses. Figure 4 provides the corresponding objective-specific loss trajectories. These values are computed on the fixed training-corpus monitor sequences at the 19 saved milestones. The total monitor loss combines the objectives using the coefficients active in each system, so its absolute magnitude is not directly comparable across systems or across objective handoffs.

For LAD Got BERT, ten late-stage gradient records yield mean weighted final-layer norms of 2.764 for RTD and 1.353 for auxiliary MLM; mean cosine similarity is 0.009. Raw records are included in `data/lad_gradient_diagnostics.csv`.

System	Vocab	Max len.	D layers	G layers	Hidden	Heads	Microbatch	Accum.
bgb-decay	16K	128	12	4	384	6	8	16
Baby Got BERT	16K	128	12	4	384	6	8	16
Lilliput Lexicon	16K	128	12	4	384	6	8	16
Lilliput Curriculum	16K	128	12	4	384	6	8	16
LAD Got BERT	16K	128	12	4	384	6	8	16

Table 4: Architecture and batching for the executed runs. All tokenizers are 16K SentencePiece unigram models trained on the permitted corpus.

System	Joint LR	MLM-only LR	Weight decay	Base mask rate	Additional details
bgb-decay	5×10^{-4}	–	0.01	0.15	RTD 50 for epochs 1–4, 25 for 5–7, 10 for 8–10
Baby Got BERT	5×10^{-4}	–	0.01	0.15	RTD 50 throughout all 10 epochs
Lilliput Lexicon	5×10^{-4}	2×10^{-4}	0.01	0.15	Adaptive uniform mix 0.30; WWM target 0.20→0.15
Lilliput Curriculum	5×10^{-4}	2×10^{-4}	0.01	0.15	Adaptive mix 0.30; WWM target 0.18→0.15
LAD Got BERT	5×10^{-4}	–	0.01	0.15	Adaptive auxiliary masks from epoch 6; update every 100 steps; uniform exploration 0.16

Table 5: Optimization hyperparameters. All runs use seed 42. Joint phases use generator and auxiliary discriminator MLM coefficients of one. Complete phase schedules are merged into Table 1 in the main paper.

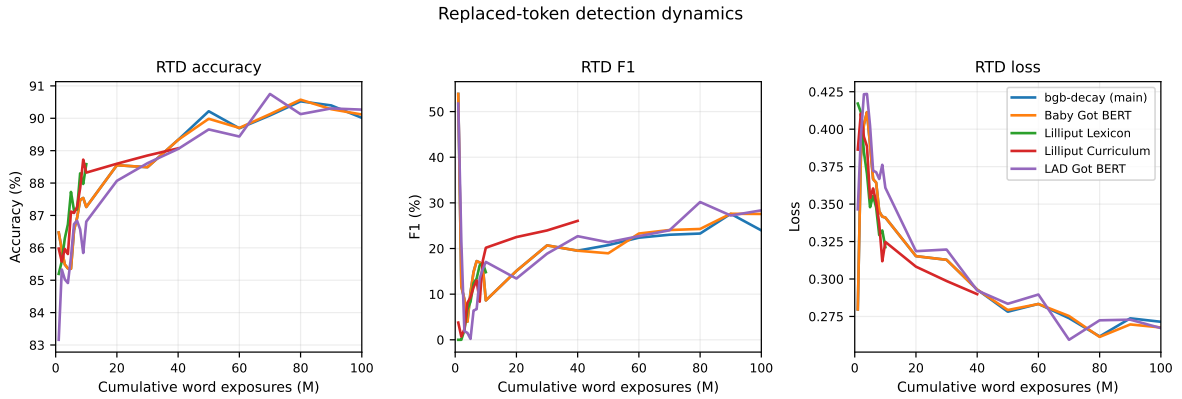


Figure 2: Replaced-token detection monitor trajectories. Lines end when RTD is disabled; an inactive objective is not plotted as zero performance. Accuracy is high partly because original tokens are the majority class, so F1 and loss are shown alongside it.

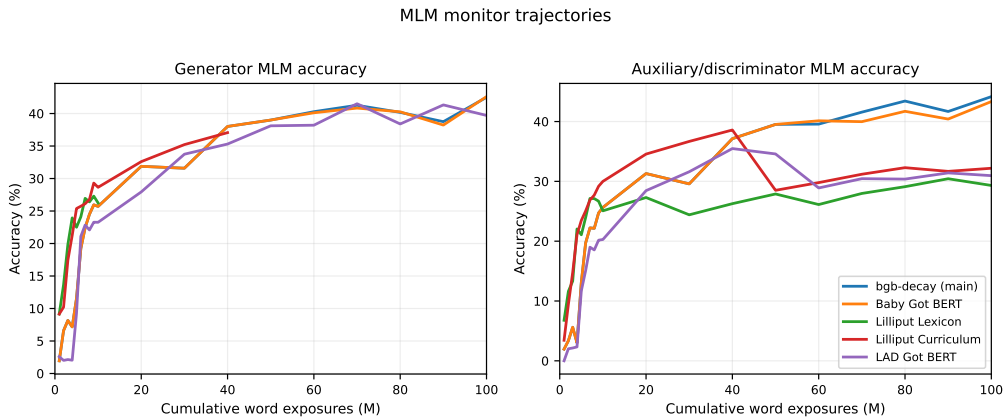


Figure 3: Generator and auxiliary/discriminator MLM accuracy on fixed training-corpus monitor sequences.

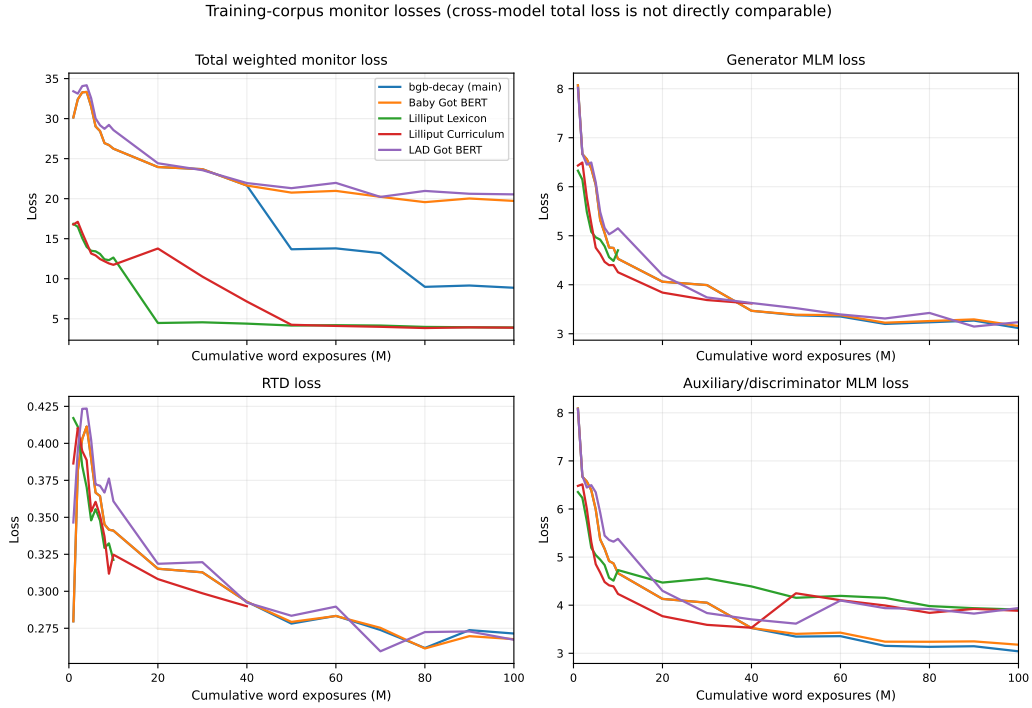


Figure 4: Checkpoint-level training-corpus monitor losses parsed from the 19 saved milestones for each model. Lines connect the recorded checkpoint values without smoothing. Zero-valued generator or RTD records are omitted after the corresponding objective is disabled. Because the systems differ in active objectives and loss coefficients, the total weighted monitor loss should not be compared directly across all models.

Final available checkpoint	Exposure	Aux MLM acc.	G MLM acc.	RTD acc.	RTD F1	RTD weight
bgb-decay	100M	0.4416	0.4249	0.9001	0.2397	10
Baby Got BERT	100M	0.4334	0.4260	0.9012	0.2757	50
Lilliput Lexicon	100M	0.2932	—	—	—	0
Lilliput Curriculum	100M	0.3219	—	—	—	0
LAD Got BERT	100M	0.3095	0.3971	0.9027	0.2836	50

Table 6: Final monitor diagnostics parsed from the supplied logs. Dashes denote objectives inactive at the final checkpoint. These are training-corpus diagnostics, not held-out scores.

E Checkpoint-Level Fast-Evaluation Diagnostics

The validated submission JSON files contain fast-subset results for all 19 milestones, from 1M through 100M cumulative word exposures. Figure 5 visualizes these checkpoint-level previews. They are useful for inspecting when particular capabilities emerge during training, but the fast subsets are smaller and noisier than the final evaluation suite. Accordingly, we do not use these trajectories for model ranking, effect-size claims, or checkpoint selection.

The plotted values are category means computed from the per-subtask fast scores stored in the validated submission JSON. Fast EWoK is omitted because the supplied local fast-collation branch does not implement the canonical context-plus-target comparison. All conclusions in the main paper therefore rely on the final full-suite results in Table 3.

Fast-evaluation checkpoint trajectories from validated submission JSON

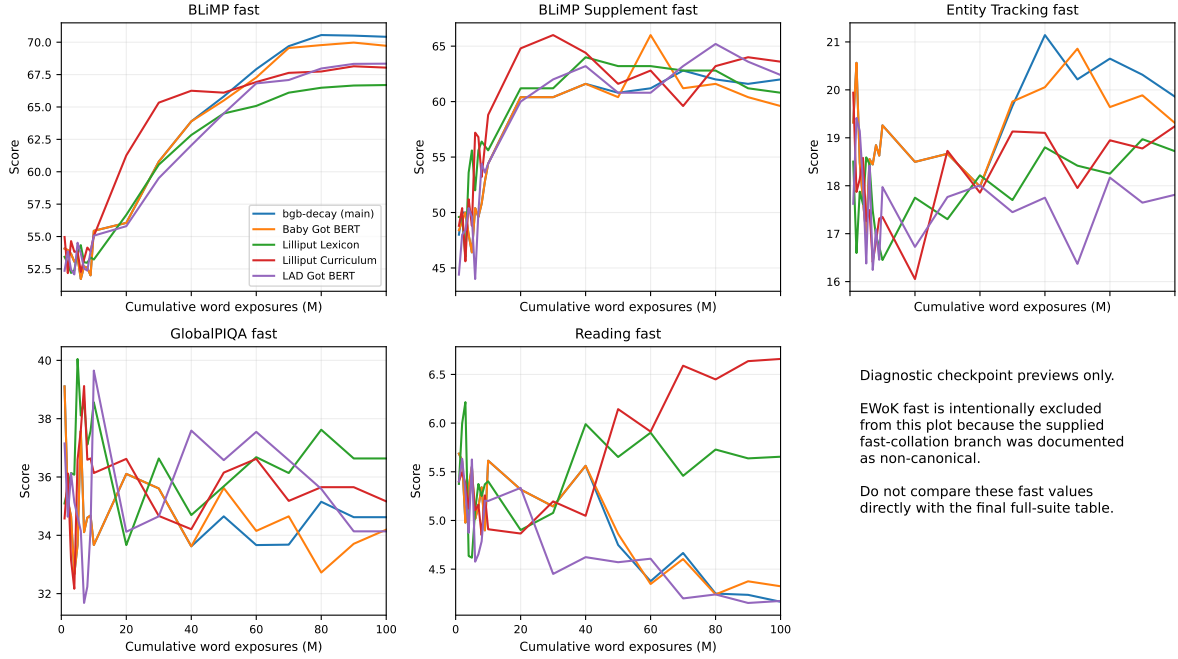


Figure 5: Diagnostic fast-evaluation trajectories across the 19 saved checkpoints. These curves show BLiMP, BLiMP Supplement, Entity Tracking, GlobalPIQA, and Reading fast-subset scores stored in the validated submission JSON. They are checkpoint previews rather than final full-suite results and should not be used for model ranking. Fast EWoK is omitted because the available local fast-collation branch is non-canonical.

F Evaluation Artifact Integrity

Using the official BabyLM 2026 evaluation implementation, the evaluation notebooks validate a native DeBERTa-v2 masked-LM artifact, check all 19 required checkpoint directories, rerun the updated Entity Tracking, GlobalPIQA, and AoA tasks, and collate the canonical result JSON. Model 5 was evaluated from scratch; Models 1–4 reused unchanged raw outputs when available and reran the updated tasks.

Several logs reach an optional local score-preview import that fails because the local utils module is not found. This occurs after successful preflight, canonical collation, and saved result JSON, so we treat it as a preview-path defect rather than a failed evaluation artifact.

The official AoA implementation changed during development, so earlier high AoA outputs are not retained. The validated submissions contain the current `aoa_surprisals` field. This distinction matters because the current leaderboard assigns AoA zero to legacy submissions that omit this field and therefore require re-evaluation. That compatibility rule does not explain the present results: all five runs were successfully collated with

Model	Loaded	Fitted	AoA
bgb-decay	504	42	0.00
Baby Got BERT	504	25	0.00
Lilliput Lexicon	504	248	0.00
Lilliput Curriculum	504	241	0.00
LAD Got BERT	504	104	0.00

Table 7: AoA fitting coverage under the current evaluator. “Loaded”/“Fitted” are CDI word counts. All runs retain more than the minimum three valid words, but their model–child correlations do not pass the evaluator’s $p \leq 0.1$ reporting threshold; counts are diagnostic coverage, not performance scores.

the new field. Instead, each run retained more than three valid fitted words but returned the failure-form diagnostic without a p -value, which identifies the evaluator branch used when the model–child AoA correlation has $p > 0.1$. Table 7 reports the resulting fitting coverage. These counts measure the number of words for which both model and child acquisition curves could be fitted and should not be interpreted as additional performance scores.